Dimensionality Reduction

An In-Depth Walkthrough of High-Dimensional Data Visualizations, Probabilistic Modeling, & t-SNE Pipeline Orchestration.

The Curse of High Dimensions

Visualization Obstacles

Human perception is bound to three spatial dimensions. Directly viewing your pipeline's 50-dimensional synthetic space is physically impossible without advanced projection modeling.

Standard scatter plots can only visualize 2D or 3D axes, ignoring the rich correlation structures hidden across the other 47 dimensions of your dataset.

Information Distortion

Taking simplistic 3D proxies (like columns 0, 15, and 45 from the code) yields heavily overlapping, messy plots with poor cluster visibility.

Euclidean distances become uniform in high-dimensional spaces, causing standard geometric clustering strategies to fail as the number of variables scales upward.

Data Synthesis & Prep Steps

1. Blob Synthesis

Generates 500 records across 50 dimensions using Scikit-Learn's `make_blobs`. Focuses on producing 5 distinct centroid clusters.

2. Scaling Features

Employs `StandardScaler` to yield a mean of zero and standard deviation of one. This guarantees uniform impact across all fifty variables.

3. Proxy Plotting

Isolates indices [0, 15, 45] for a raw 3D scatter. This showcases the immediate visual overlap prior to applying structural optimization.

The Core Concept of t-SNE

Preserving Neighborhood Topologies

Unlike standard projections that maximize variance global structures (like PCA), t-SNE focuses on preserving small local variances.

- Converts pairwise spatial metrics into probabilistic similarities.
- Keeps neighbors close while pushing non-similar groups apart.
- Utilizes non-linear mapping capabilities to unravel complex, highly curved 50-dimensional manifolds into clean 2D components.

t-SNE clusters visual showing colorful separated distributions

High-D Similarity Mapping

1. Gaussian Probabilities

In the original 50D space, the distance from point j to point i is mapped to a conditional probability under a Gaussian distribution centered at i:

$$p_{j|i} = \frac{\exp - \frac{\| x_i - x_j \|^2}{2\sigma_i^2}}{\sum_{k \neq i} \exp - \frac{\| x_i - x_k \|^2}{2\sigma_i^2}}$$

2. Symmetric Symmetrization

To reduce sensitivity to outlier anomalies, conditional metrics are averaged into a symmetric joint probability matrix:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$$

Low-D Similarity Mapping

Student-t Kernel (1 DOF)

For mapping the points in compressed 2D space, a heavy-tailed Student-t distribution with 1 degree of freedom (equivalent to a Cauchy kernel) is chosen:

$$q_{ij} = \frac{\left(1 + \|y_i - y_j\|^2\right)^{-1}}{\sum_k \sum_{l \neq k} \left(1 + \|y_k - y_l\|^2\right)^{-1}}$$

Mitigating the Crowding Problem

The volume of space drops rapidly from 50D to 2D. Under normal distribution mapping, points on moderate high-dimensional bounds compress into a single congested mass.

The long tails of the Student-t distribution allow modelers to space apart clusters while accurately retaining micro-relationships.

KL Divergence Optimization

- **The Loss Objective:** Minimizes discrepancies between target distributions (P) and embedding similarities (Q).
- **Asymmetric Behavior:** Large penalizations for bringing small-distance High-D coordinates far apart in Low-D space.
- **Iterative Refinement:** Optimized utilizing gradient descent configurations via internal parameter calculations.

The Kullback-Leibler (KL) Cost Function:

$$C = KL(P \,||\, Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

t-SNE Pipeline Parameters

Parameter	Value In Code	Target Behavior	Impact of Incorrect Tuning
n_components	2	Defines dimensional output size (2D target projection).	Greater dimensions complicate manual plotting and review.
perplexity	35	Controls effective neighbors size; adjusts local scale variance.	Oversized values merge clusters; small scales split groups.
max_iter	1200	Maximum optimization iteration steps.	Insufficient limits end mapping processes prematurely.
learning_rate	'auto'	Gradient step speed during coordinates optimization.	High values disrupt gradients; low steps cause lock-ups.

Reduction Performance Comparison

Metric evaluation of cluster purity scores reveals that linear projections (like PCA) struggle with 50D overlap. Your pipeline's t-SNE algorithm resolves non-linear boundary constraints with optimal visual separation.

Questions?

Your t-SNE dimensionality pipeline execution is complete.

File Generated: high_dim_to_low_dim_tsne.png | scikit-learn Pipeline

Image Sources

Thumbnail for la.mathworks.com https://la.mathworks.com/help/examples/stats/win64/VisualizeHighDimensionalDataUsingTSNEExample_01.png

Source: la.mathworks.com